

Package ‘leafblower’

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Title High-Performance Survey Calibration via iEPPA and L-BFGS-B

Version 0.1.0

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Description Drop-in replacement for `autumn::harvest()` implementing iEPPA (Sinkhorn block-coordinate descent) and L-BFGS-B over the Deville-Sarndal logit dual. Adds `min_weight` lower bound; shares a single C++17 core with the Python bindings.

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Encoding UTF-8

NeedsCompilation yes

SystemRequirements C++17 compiler, OpenMP (optional, for SIMD exp vectorisation)

Suggests `autumn` (>= 0.2.0), `testthat` (>= 3.0.0), `bench`, `lhs`, `DiceKriging`, `ggplot2`, `rprojroot`

RoxygenNote 7.3.3

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anesrake

anesrake compatibility wrapper

Description

Maps anesrake parameter names to harvest(). Unknown params produce warnings.

Usage

```
anesrake (
  inputter,
  targets,
  weightvec = NULL,
  caseid = NULL,
  pctlim = 0.05,
  cap = 5,
  choosemethod = "rake",
  type = "pctlim",
  nlim = 500L,
  iterate = TRUE,
  threads = 1L,
  ...
)
```

Arguments

inputter	Data frame (maps to data).
targets	Named list of target proportions.
weightvec	Starting weights or NULL.
caseid	Ignored (row label not used by calibration).
pctlim	Deprecated; maps to <code>convergence["pct"]</code> .
cap	Upper weight cap; maps to <code>max_weight</code> .
choosemethod	"rake" or "nrake"; maps to <code>method="lbfgsb"</code> with warning.
type	Ignored.
nlim	Max iterations; maps to <code>max_iterations</code> .
iterate	Ignored (always iterates).
threads	Ignored (single-threaded v1).
...	Unknown params produce a warning then are ignored.

Value

Same as `harvest()`.

design_effect	<i>Kish design effect (1-argument) or Henry-Valliant (4-argument)</i>
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Description

When called with only `weights`, computes the Kish (1992) design effect: $n * \sum(w^2) / \sum(w)^2$.

Usage

```
design_effect(weights, outcome = NULL, data = NULL, target = NULL)
```

Arguments

<code>weights</code>	Numeric vector of calibrated weights.
<code>outcome</code>	Numeric outcome vector (optional; 4-arg form only).
<code>data</code>	Data frame used in calibration (optional; 4-arg form only).
<code>target</code>	Named list of target proportions (optional; 4-arg form only).

Details

When called with all four arguments, computes the Henry & Valliant (2015) calibration design effect: `var_weighted(y) / var_unweighted(y)`.

Value

Numeric scalar design effect.

diagnose_weights	<i>Diagnose calibration quality</i>
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Description

Returns a data frame comparing original and weighted marginals to targets.

Usage

```
diagnose_weights(data, target, weights)
```

Arguments

<code>data</code>	Data frame used in calibration.
<code>target</code>	Named list of target proportions (same format as <code>harvest()</code>).
<code>weights</code>	Numeric vector of calibrated weights, length <code>nrow(data)</code> .

Value

Data frame with columns: `variable`, `level`, `prop_original`, `prop_weighted`, `target`, `error_original`, `error_weighted`.

effective_sample_size
Effective sample size

Description

Effective sample size

Usage

```
effective_sample_size(weights)
```

Arguments

`weights` Numeric vector of calibrated weights.

Value

Numeric scalar: $n / \text{design_effect}(\text{weights})$.

get_current_miss *Current calibration miss (matching autumn's export name)*

Description

Current calibration miss (matching autumn's export name)

Usage

```
get_current_miss(data, target, weights)
```

Arguments

`data` Data frame.
`target` Named list of target proportions.
`weights` Numeric weight vector.

Value

Named numeric vector of max absolute errors per variable.

harvest

Generate calibrated weights (drop-in for autumn::harvest)

Description

Generate calibrated weights (drop-in for autumn::harvest)

Usage

```
harvest(
  data,
  target,
  min_weight = 0,
  max_weight = 5,
  method = "ieppa",
  verbose = 0,
  max_iterations = 500,
  start_weights = NULL,
  attach_weights = TRUE,
  weight_column = "weights",
  convergence = list(),
  select_params = NULL,
  select_function = NULL,
  error_function = NULL,
  adaptive_order = NULL,
  enforce_mean = TRUE,
  accelerate = FALSE,
  add_na_proportion = FALSE,
  auto_collapse = FALSE,
  collapse_vars = NULL,
  target_map = NULL,
  ...
)
```

Arguments

data	A data frame containing columns to calibrate.
target	A named list of named numeric vectors (variable -> proportions).
min_weight	Lower bound on weights. Default 0 (no lower bound).
max_weight	Upper bound on weights. Default 5.
method	One of "auto", "ieppa", "lbfgsb", "raking". "ieppa": paper-faithful algBCD at C=0 (cell-compressed Sinkhorn, Chu-Liang-Toh-Yang 2022 arXiv:2011.14312). "raking": classical IPF + Dykstra box + hyperplane projections (Deming-Stephan 1940 / Csiszar 1975 + Boyle-Dykstra 1986). Renamed from the prior misnamed "iEPPA" hybrid. "lbfgsb": L-BFGS-B on the Deville-Sarndal logit dual. "auto" (default): currently routes unconditionally to "ieppa". Benchmark-driven routing refinement is planned.

verbose	Integer verbosity: 0=silent, 1=progress, 2=debug.
max_iterations	Maximum inner BCD iterations per outer step. Default 500.
start_weights	Starting weights vector or NULL (uniform).
attach_weights	If TRUE, return data frame with weights column. Default TRUE.
weight_column	Name of weight column. Default "weights".
convergence	Named list/vector; "absolute" maps to tol_abs.
select_params	Ignored with verbose >= 2 note.
select_function	Ignored.
error_function	Ignored.
adaptive_order	Ignored.
enforce_mean	Ignored (retained for compatibility).
accelerate	Ignored.
add_na_proportion	Not supported in v1; raises error if TRUE.
auto_collapse	Not supported in v1; raises error if TRUE.
collapse_vars	Not supported in v1; raises error if TRUE.
target_map	Passed through for data-frame target format handling.
...	Additional arguments ignored.

Value

data frame with weights column if attach_weights=TRUE, else numeric vector.

weighted_pct	<i>Weighted proportions</i>
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Description

Weighted proportions

Usage

```
weighted_pct(x, weights)
```

Arguments

x	Factor or character vector.
weights	Numeric weights, same length as x.

weighted_pct

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Value

Named numeric vector of weighted proportions summing to 1.

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